

FM Demodulation and Mono Audio Recovery

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I. INTRODUCTION

In this project, a down-converted FM broadcast signal is provided for analysis. The sampling frequency of the signal is 240 kHz, and the target audio sampling frequency is 48 kHz.

The objective is to analyze the received signal, demodulate it, isolate the mono audio component, and reconstruct the audio signal.

An FM signal can be expressed as[1]

$$x(t) = A \cos\left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau) d\tau\right), \quad (1)$$

where $m(t)$ is the message signal and k_f is the frequency deviation constant.

II. RECEIVED SIGNAL SPECTRUM

The amplitude spectrum of the received FM signal is examined first to identify the occupied bandwidth.

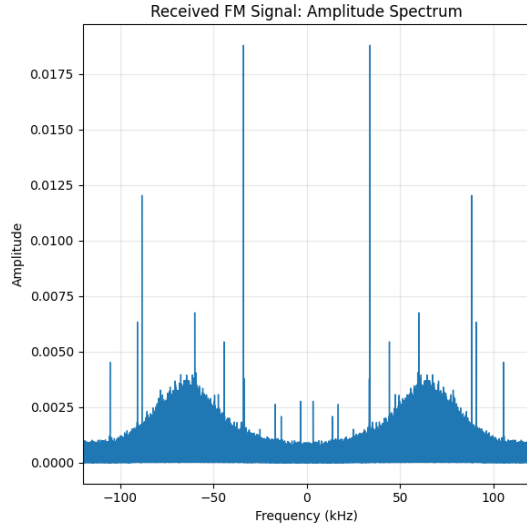


Fig. 1: Amplitude spectrum of the received FM signal sampled at 240 kHz.

The wide bandwidth observed in the spectrum is consistent with FM modulation, where the information is encoded in frequency variations[1].

III. FM DEMODULATION

FM demodulation is performed using the analytic signal and phase differentiation method. The analytic signal is obtained using the Hilbert transform[1]:

$$x_a(t) = x(t) + j \mathcal{H}\{x(t)\} = A e^{j\phi(t)}, \quad (2)$$

where $\phi(t)$ denotes the instantaneous phase.

The instantaneous frequency is then calculated as the time derivative of the phase[1]:

$$f_{\text{inst}}(t) = \frac{1}{2\pi} \frac{d\phi(t)}{dt}. \quad (3)$$

In discrete time, this is approximated by

$$f_{\text{inst}}[n] = \frac{f_s}{2\pi} (\phi[n] - \phi[n-1]), \quad (4)$$

where f_s is the sampling frequency. The DC component corresponding to the carrier frequency is removed to focus on the message content.

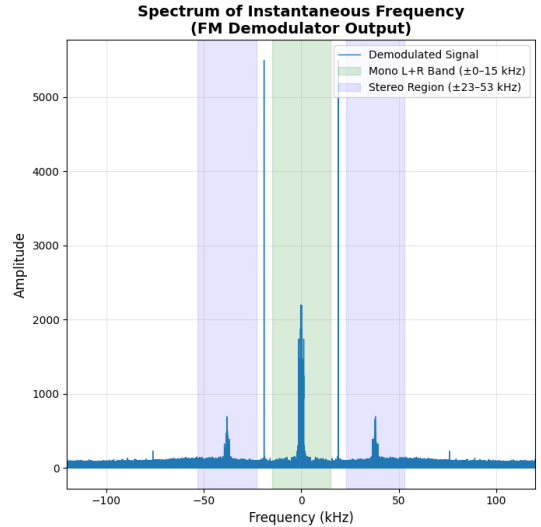


Fig. 2: Spectrum of the instantaneous frequency after FM demodulation.

The mono L+R band is visible in the 0–15 kHz region. The mono audio information is clearly concentrated at low frequencies.

IV. MONO EXTRACTION

A low-pass FIR filter with a cutoff frequency of 15 kHz is applied to isolate the mono component[3]:

$$m_{\text{mono}}[n] = f_{\text{inst}}[n] * h_{\text{LPF}}[n], \quad (5)$$

where $h_{\text{LPF}}[n]$ is the impulse response of the low-pass filter.

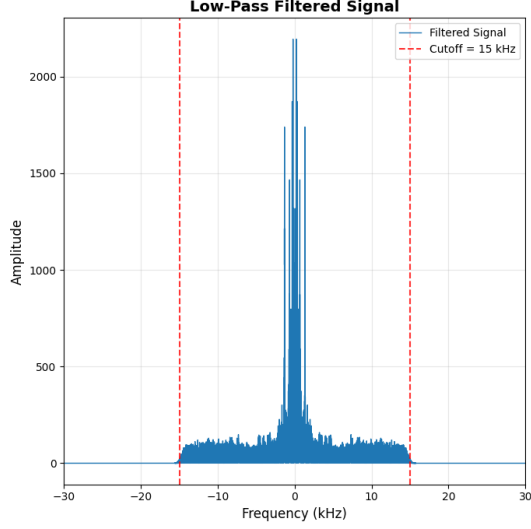


Fig. 3: Spectrum after low-pass filtering the demodulated signal. Only the mono L+R band remains after filtering.

This step removes higher-frequency multiplex components.

V. DOWNSAMPLING AND DE-EMPHASIS

After filtering, the signal is downsampled by an integer factor of five to reach the target sampling rate.

Then, a de-emphasis filter with a time constant of $75 \mu\text{s}$ is applied. This filter is implemented using a bilinear transformation.[3]

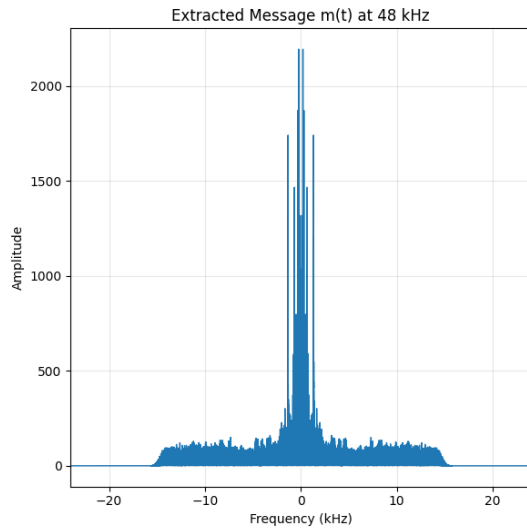


Fig. 4: Spectrum of the extracted message signal at 48 kHz before de-emphasis.

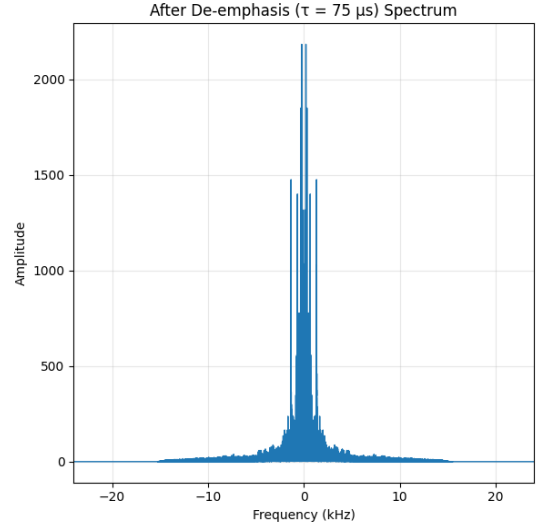


Fig. 5: Spectrum after applying the de-emphasis filter. High-frequency components are attenuated to restore audio balance.

VI. FINAL AUDIO SIGNAL

The signal is normalized to prevent clipping and saved as a WAV file:

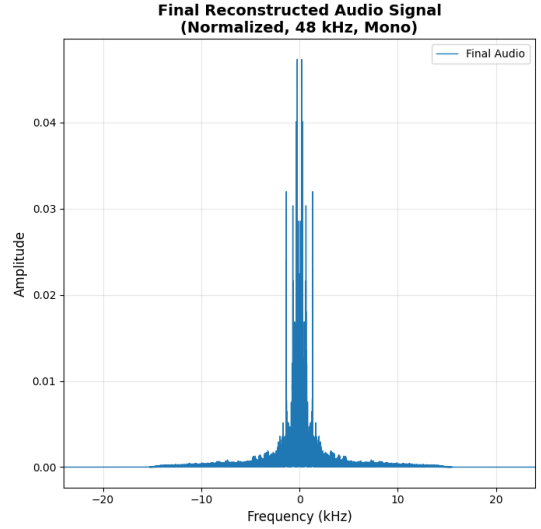


Fig. 6: Spectrum of the final reconstructed mono audio signal at 48 kHz.

The recovered audio is clear and matches the expected mono broadcast signal.

REFERENCES

- [1] Ali Emre Pusane, EE371, 2025-2026 Lecture Notes
- [2] B. P. Lathi and Z. Ding, *Modern Digital and Analog Communication Systems*, 4th ed. New York, NY, USA: Oxford University Press, 2009.
- [3] J. G. Proakis and D. K. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th ed. Upper Saddle River, NJ, USA: Pearson, 2007.